



TeaNet8: A real time Android application-based Tea Leaf Disease detection using fine-tuned transfer learning and Gradient-Weighted Class Activation Mapping visualization

Ismotara Dipty^{a,e}, Md Assaduzzaman^{a,e,*},
Nafiz Fahad^{b,d}, Md. Jakir Hossen^c,
Md. Farhatul Haider^a, Fiaj Rahman^a

^a Department of Computer Science Engineering, Daffodil International University, Dhaka, Bangladesh

^b Faculty of Information Science and Technology (FIST), Multimedia University, Melaka, 75450, Malaysia

^c Center for Advanced Analytics (CAA), COE for Artificial Intelligence, Faculty of Engineering & Technology (FET), Melaka, 75450, Malaysia

^d EliteLab.Ai, New York, United States of America

^e DeepHealth Research Lab, Dhaka, Bangladesh

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ABSTRACT

Tea is widely regarded as one of the most popular beverages globally, and Bangladesh plays a significant role both as a producer and consumer of this renowned drink. However, diseases that impact the quality and productivity of crops can greatly impede the production of tea, impacting the final product's quantity and quality. To prevent and control tea leaf diseases, a reliable and precise diagnosis and identification system is needed. Tea leaf infections are discovered manually, which takes time and affects crop quality and production. Detecting tea leaf disease early can lead to decreased damage to overall tea production. Advanced deep learning methods are simplifying the identification and categorization of specific illnesses in tea plants. This current study introduces TeaNet8, a deep learning-based approach for identifying and classifying eight tea leaf disease classes using a fine-tuned ResNet50V2 model. Moreover, this study employs 2824 images of eight different types of leaf diseases. Preprocessing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE), brightness adjustment, and unsharp masking were applied to enhance the dataset. Additionally, data augmentation techniques were used to increase its diversity. The proposed model identify the different type of tea leaf disease with 97% accuracy. Gradient-Weighted Class Activation Mapping (Grad-CAM) visualization was employed to interpret and understand model predictions. The model demonstrated perfect accuracy for Algal Spot, Anthracnose, Gray Blight, and White Spot, with accuracy rates of 97.14% for Brown Blight, 94.59% for Healthy leaves, 94.12% for Red Spot, and 92.31% for Bird Eye Spot. Furthermore, the proposed model's performance was compared against three pre-trained fine-tuning models. Various performance measurement indicators were used to evaluate the performance of the proposed model. The results showed that the proposed model is effective in categorizing diseases in tea leaves. Finally, An Android-based system was developed employing the most effective model to aid farmers for detecting tea leaf diseases.

* Corresponding authors.

E-mail addresses: assaduzzaman.cse@diu.edu.bd (M. Assaduzzaman), jakir.hossen@mmu.edu.my (M.J. Hossen).

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1. Introduction

Tea, the beverage that is second in popularity globally only after water, is derived from the leaves and buds of the “*Camellia Sinensis* Plant”. Tea is distinguished by its special qualities due to its specific growth in particular environments and weather conditions [1]. Tea is a popular beverage globally because of its enjoyable taste, pleasant flavor, and health advantages. It contains numerous potent phytochemicals that provide significant health benefits to humans. Compared to other plant species, tea grows rather widely, almost on every hill. Around the world, tea significantly contributes to the strengthening of family and friend ties. Bangladesh is a major tea-producing nation. Tea is one of Bangladesh’s most productive agro-industrial and export crops. Most people consume it frequently, and people like it both inside and outside of its country of origin. The roots of Bangladesh’s tea industry trace back to the British colonial period, when the East India Company initiated tea cultivation in the Sylhet region’s hills. Subsequently, tea cultivation expanded to Greater Chittagong in 1840, further solidifying the industry’s foundation. In 2021, Bangladesh was ranked as the 66th largest tea exporter globally, with tea exports amounting to \$2.77 million. In the same year, tea was the 192nd highest exported item from Bangladesh [2]. The FAO reports that the global tea industry is worth approximately \$9.5 billion, with global tea production surpassing \$17 billion [3]. Today, Bangladesh is home to 166 commercial tea estates, which include some of the world’s largest working plantations. The tea industry contributes to 3% of global tea production and serves as a vital economic sector, employing over 4 million people. This thriving industry not only bolsters the country’s economy but also highlights Bangladesh’s longstanding tradition of producing high-quality tea that is cherished both locally and globally.

However, the quality and productivity of the crop are significantly impacted by diseases brought on by fungi, bacteria, viruses, and environmental variables, which significantly challenge tea cultivation. Farmers traditionally find tea leaf infections through microscopic analysis or visual examinations. These procedures take time, are prone to error, and often subjective. More effective and scalable methods for real-time detection and treatment of tea plant diseases are vital as their frequency increases.

To address this challenge, we have developed a real-time Android application that utilizes advancements in deep learning, image processing techniques, and Gradient-weighted Class Activation Mapping (Grad-CAM) to facilitate automated and interpretable disease detection. Although these technologies have demonstrated potential in agriculture, their application for real-time solutions, particularly in tea cultivation, has been limited until now. Through the integration of Grad-CAM, our application enhances disease detection by delivering visual explanations by highlighting the areas of tea leaves most indicative of the disease, offering farmers more detail transparency and insight into the diagnosis process. The adoption of real-time applications in tea farming has the potential to revolutionize practices in Bangladesh, bringing measurable economic benefits to the industry. Early detection of tea leaf diseases, enabled by cutting-edge technology, can help farmers identify and address issues before they escalate, significantly reducing crop losses. This proactive approach minimizes reliance on costly chemical treatments and ensures healthier crops, translating into higher yields and better-quality tea. Moreover, integrating real-time monitoring and predictive tools can optimize resource utilization, such as water and fertilizers, reducing production costs. By leveraging such technology, farmers can make informed decisions, leading to increased profitability and sustainable farming practices. For a country like Bangladesh, where tea production plays a pivotal role in the economy, these advancements can enhance competitiveness in the global market while improving the livelihoods of millions dependent on the industry.

The remainder of this paper is organized as follows: Section 2 reviews related works on tea leaf disease detection using deep learning. Section 3 details the proposed system architecture, including dataset preprocessing, model design, and training methodology. Section 4 presents the experimental results, performance analysis, and ablation study. Finally, Section 5 concludes the study, highlighting key contributions and future directions.

1.1. Disease description

Many illnesses impacting tea plant leaves greatly influence the amount and quality of tea produced. Some typical diseases of tea leaves include red leaf spot, white spot, bird eye spot, Anthracnose, brown blight, algal leaf, and gray blight. Early detection of these diseases and application of natural remedies are important components of disease management in agriculture. Understanding the causes and consequences of plant diseases is crucial for proactive disease management in agriculture. This section will provide a full discussion of common tea leaf diseases.

1.1.1. Algal leaf

Algal leaf spot is identified by cushion-like patches on the leaf’s surface, which can appear in shades of grayish, green, brown, or orange. Tea plants can be easily affected by a plant disease known as algal leaf disease, which is also called algal leaf spot [4]. *Cephaleuros* is a parasitic green alga that lives on tea plants. Many different species of algae, primarily those belonging to the genera, *Cephaleuros* and *Asterina*, are the cause of this disease. Warm, humid weather that promotes the growth of algae in areas that grow tea is the main cause of algal leaf disease. The appearance of green to dark green patches or spots on the upper surface of tea leaves is the main sign of algal leaf disease.

1.1.2. Anthracnose

Anthracnose (*Gloeosporium theae-sinensis* Anthracnose or miyake) is a dangerous foliar disease that harms tea plants and puts tea production and quality at risk [5]. *Colletotrichum* is the primary cause of anthracnose, a dangerous leaf disease that has severely reduced tea yield and quality. It explains the damage that different fungal diseases bring to plants. Dangerous fungus known as anthracnose targets a wide variety of plant species.

1.1.3. Bird eye spot

Bird eye spot disease can affect tea bushes of any age, but young tea plants in nurseries and fields are particularly susceptible [6]. The upkeep and tea young leaves made it clear that the disease was widespread in southwest Ethiopia. The fungus responsible for this illness is called *Botryosphaeria dothidea*. The fruit may thus acquire microscopic, dark, sunken lesions as a result, which frequently resemble bird's eyes and give rise to the term "Bird's Eye Spot". The symptoms manifest as round, gray-brown, 10 mm-diameter spots on the leaves with a dot of a lighter color in the center.

1.1.4. Brown blight

Brown blight *Colletotrichum camelliae* was a very common leaf disease and occurred in every tea field [7]. But since it has nothing to do with immature leaves that can be harvested, there is no monetary loss. The first indications of it are shrubs that have been weakened or damaged by waterlogging, diseases of the stem and roots, severe pruning, hail, frost, sunburn, or herbicide damage. Lilian Blight in the Field Symptoms Green-yellow patches appeared on the young leaves. Mycelium that is red and dray appears as irregular, round patches on the leaves. These diseases afflict older leaves.

1.1.5. Gray blight

Grey blight, a common fungal disease caused by a species similar to *Pestalotiopsis*, affects tea crops in many countries where tea is grown. It typically causes gray to light brown lesions on the leaves [8]. Eventually, these lesions may result in leaf drop and a lowering of the tea's quality. They may be shaped asymmetrically or angularly. Grey blight is caused by *Cercospora*. In addition to brown blight, there is another extremely prevalent leaf disease. It afflicted weak bushes, as well as those injured by hail, frost, sunburn, hard plucking, herbicide damage, and diseases of the stem and roots, similar to brown blight.

1.1.6. Red leaf spot

The disease known as "red leaf spot" affects creeping bentgrass in the spring, summer, or fall when the weather is warm and wet. Red leaf spot is a "Helminthosporium" brought on by fungi that produce big, cigar-shaped spores. One such disease is red leaf spot [9]. Tea leaves with tiny, reddish-brown to black spots or lesions are indicative of the fungal disease known as red leaf spot. These patches may be erratic and, in the right circumstances, may clump together. Severe infections can impair the quality of the tea leaves, weaken the tea plant, and reduce yield.

1.1.7. White spot

A fungal pathogen called "white spot" can cause "Phyllosticta leaf spot" or "Phyllosticta theaifolia leaf spot", which is a disease that affects tea plants (*Camellia sinensis*). The leaves of tea plants are the primary target of this fungal pathogen [10].

1.1.8. Healthy leaf

A healthy tea leaf is vibrant green, firm, and resilient. It has a smooth texture without any spots, discoloration, or wilting. The leaf's edges are smooth and intact, and it boasts a fresh, invigorating aroma.

1.2. Contribution

This study presents the following key contributions:

- Employed several image pre-processing techniques such as Contrast Limited Adaptive Histogram Equalization, brightness adjustment, and unsharp masking to improve image quality, along with augmentation methods to enhance model accuracy.
- Investigated the relationship between model performance, data augmentation techniques, and ablation studies to enhance the classification accuracy of tea leaf diseases.
- Implemented Grad-CAM to visually highlight critical regions on tea leaves, aiding cultivators in better understanding disease indicators.
- Developed a real-time Android application to assist farmers in the early detection of tea leaf diseases.

2. Related works

This section examines and evaluates various research and literature on detecting diseases in tea leaves. Recent studies highlight technological breakthroughs, including advancements in resources and technology. The literature explores a range of methodologies and approaches contributing to the progress in tea leaf disease detection.

Saikat Datta et al. (2023) utilize a Complex Convolutional Neural Network with numerous hidden layers for accurately classifying unhealthy tea leaves into different categories [11]. The accuracy of disease detection is improved by the network's ability to detect a higher number of features through the use of multiple hidden layers. Six categories are included in the classification: Red Spot, Helopeltis, Brown Blight, Algal Spot, Gray Blight, and Healthy Leaves. A labeled dataset with 5867 photos of tea leaves — both healthy and diseased — has been made and uploaded to Kaggle. The suggested method achieves an impressive 96.56% overall accuracy, according to the results. The model has potential to be helpful in managing diseases by accurately detecting various long-lasting tea leaf illnesses.

Md. JanibulAlamSoeb et al. [2023] utilizes a model to address the need for a trustworthy tea leaf disease detection system in order to improve prevention and management [12]. Labor-intensive manual detection techniques have a detrimental effect on yield quality. The researchers utilized YOLOv7, a fast, one-step model for detecting objects, to train on 4000 pictures of five various tea leaf illnesses from renowned Bangladeshi gardens. The YOLOv7 method yields impressive results with accuracy 97.3%. Comparative analysis demonstrates that YOLOv7 outperforms other detection and identification networks, like YOLOv5, through achieving a greater level of precision. Based on the research results, YOLOv7 provides an efficient way to quickly and accurately detect diseases in tea leaves, potentially decreasing entomologists' workload and financial losses in the tea sector.

Nadide Yucel et al. (2023) developed a model for plant diseases on crop quantity and quality, focusing on tea leaves. Recognizing the challenges farmers face in visually detecting diseases, a computer-aided model is developed using three Convolutional Neural Network (CNN) architectures [13]. Using these CNN architectures, the model captures feature maps from the dataset images. The effectiveness of the suggested model is demonstrated by comparing its performance with seven other CNN architectures used in previous studies. Assessment criteria show a high accuracy rate of 91.3%, demonstrating the model's potential usefulness in identifying diseases in tea plants.

Eliana Aida Rashida et al. (2023) have used technological advancements to control tea leaf diseases, focusing on four disease samples [14]. In applications of image recognition, Convolutional Neural Networks (CNN) play a vital role in classifying and managing diseases. With a dataset ratio of 70:30, 150 epochs, augmentation, precision, and recall in pre-processing, the testing procedure yields 95% accuracy, 95% precision, and 24% recall. 400 images make up the dataset, which is categorized into four groups: healthy, anthracnose, bird eye spot, and algae leaf. These findings highlight the crucial factors influencing the effectiveness of the model.

Nishmitha et al. (2023) utilizes CNN technology and SVM for precise identification and detection of diseases affecting tea leaves [15]. Image preprocessing techniques, such as segmentation and data enhancement, are applied to optimize the training dataset. The CNN model successfully classified diseases with a remarkable 92% accuracy, whereas the support vector machine achieved 88.9%. Additionally, strategies such as adjusting learning rates, varying iteration numbers, and incorporating dropout were implemented to enhance CNN's recognition accuracy.

Meenakshi Aggarwal et al. (2023) introduced [16] a federated transfer learning (F-TL) approach for rice-leaf disease classification, achieving 99% accuracy with EfficientNetB3 and 0.1 loss for both IID and non-IID datasets. MobileNetV2 showed slightly lower accuracy, with 98% (IID) and 90% (non-IID). The F-TL framework outperformed traditional distributed models, demonstrating strong performance in both single and multiclient scenarios.

Meenakshi Aggarwal (2023) et al. proposed a [17] deep learning method for classifying rice leaf diseases, comparing machine learning classifiers with CNN and transfer learning models. InceptionResNetV2 achieved the highest validation accuracy of 88%, showing that transfer learning models outperform traditional classifiers.

Meenakshi Aggarwal et al. (2022) developed [18] a federated learning-based approach for diagnosing rice leaf diseases using EfficientNetB3, achieving 99% accuracy on both IID and non-IID datasets. In comparison, traditional models like random forest and k-nearest neighbors showed lower accuracy rates of 85.42% and 78.16%, respectively. This approach ensures data privacy while outperforming conventional models, making it a promising solution for early rice leaf disease detection.

Meenakshi Aggarwal et al. (2022). [19] explored the use of advanced technologies, including Machine Learning (ML), Deep Learning (DL), and the Internet of Things (IoT), for the early detection of rice leaf diseases. Their work emphasizes the importance of leveraging these techniques to improve disease diagnosis and mitigate the impact of plant diseases on agricultural productivity. Latha et al. (2021) classified tea leaf diseases using a CNN architecture. Of the 8-class dataset, which had 860 images for training and 172 images for testing, 80% (688 images) were used in the study. The accuracy value of the model utilized in the study was 94.45%. Researchers found that as the epoch value rose in the study, the suggested model's accuracy increased as well. As a result, they said that determining the most suitable epoch value is crucial [20].

Somnath Mukhopadhyay et al. (2021) introduce a new method to identify tea leaf diseases at an early stage, aiming to minimize global crop losses due to pathogen exposure [21]. The recommended approach utilizes the Non-dominated Sorting Genetic Algorithm (NSGA-II) to cluster images and identify disease spots on tea leaves. Following principal component analysis (PCA) to reduce features, multiple-class support vector machines (SVM) are used for diagnosing diseases. The program can correctly detect five different tea leaf diseases, such as Red Rust, Red Spider, Thrips, Helopeltis, and Sunlight Scorching, with an average precision rate of 83%. This automatic technology has the potential to reduce crop losses by making the early and precise diagnosis of diseases affecting tea leaves easier.

Shulin Meng et al. (2020) proposed a technique to detect Tea Red Leaf Spot and Tea Red Scab illnesses by employing a hybrid feature optimization algorithm. The Histogram of Oriented Gradient and the Inception v3 model are employed for extracting image features, which are then enhanced through hybrid processing methods [22]. The Gradient Boosting Decision Tree algorithm serves as the classifier for identifying illnesses. Experimental results demonstrate that the hybrid feature optimization algorithm effectively reduces image features from 36,068 to less than 150 dimensions, while maintaining an identification accuracy rate of over 95%. This important simplification highlights the algorithm's capability to accurately and effectively detect diseases that impact tea leaves.

Jing Chen et al. [2019] developed a deep convolutional neural network (CNN) known as LeafNet for identifying diseases in tea plant leaves using images [23]. LeafNet's feature extractor filters automatically capture features related to diseases. This research presents a model for disease classification that utilizes a dense scale-invariant feature transform (DSIFT) feature-based bag of visual words (BOVW) approach. When directly compared, LeafNet outperforms SVM and MLP classifiers, reaching a remarkable accuracy of 90.16%. On the other hand, SVM shows an accuracy of 60.62% while MLP has a slightly higher accuracy of 70.77%. The findings

Table 1
Overview of the previous work.

SL No	Authors' name	Year	Model	Accuracy
01	Saikat Datta et al.	2023	CNN	96.56%
02	Md. Janibul Alam Soeb et al.	2023	YOLOv7	97%
03	Nadide Yucel et al.	2023	CNN	91.3%
04	Eliana Aida Rashida et al.	2023	CNN	95%
05	Nishmitha et al.	2023	CNN	92%
06	Meenakshi Aggarwal et al.	2023	CNN	88%
07	Meenakshi Aggarwal et al.	2022	Machine Learning	Not mentioned
08	Latha et al.	2021	CNN	94.45%
09	Somnath Mukhopadhyay et al.	2022	SVM	83%
10	Jing Chen et al.	2020	LeafNet	90.23%
11	Shulin Meng et al.	2020	InceptionV3	95%
12	Jing Chen et al.	2019	LeafNet	90.16%
13	Xiaoxiao et al.	2018	CNN	93.75%

demonstrate the efficiency of LeafNet in detecting tea leaf diseases and its potential to enhance the precision of disease diagnosis in upcoming assessments.

Xiaoxiao et al. (2018) utilized a model based on convolutional neural networks to safeguard tea leaves against disease. The data was inputted to the network after it underwent data augmentation and image segmentation. The frequency of the iterations in the CNN architecture has been altered by the learning rate in order to increase recognition accuracy. On the other hand, the accuracy of the SVM classifier in identifying CNN-related diseases was 93.75% [24].

Recent studies, such as Learning From AI-Generated [25] Annotations for Medical Image Segmentation, demonstrate the potential of AI in automating annotation tasks. These advancements align with our use of transfer learning and data augmentation to improve model robustness. Recent advancements in neural networks have expanded their applications across various domains. For example, novel procedures have been developed for solving complex systems like singular fifth-order nonlinear multi-pantograph differential equations [26] and fractional economic models using Bayesian regularization radial basis neural networks [27]. Neural networks have also been applied to model the Zika virus spread [28], simulate robot-based coronavirus detection with Levenberg–Marquardt backpropagation [29], and study hepatitis B virus dynamics with hyperbolic tangent sigmoid deep networks [30]. Additionally, combinations of radial basis and sigmoid networks have addressed HIV in cancer patients [31], while reliable methods have tackled sixth-order nonlinear pantograph models [32]. Scale conjugate gradient networks predict hard water consumption based on kidney function, and stochastic networks handle fifth-order singular engineering models [33]. These studies highlight the growing versatility of neural networks in solving intricate problems. These studies underscore the growing importance of neural networks in addressing complex challenges.

This research sets the foundation for upcoming efforts to tackle the current lack of classification for tea leaf diseases. The research utilized various Convolutional Neural Network (CNN) models that utilize transfer learning and introduced a novel model designed specifically for classifying diseases that impact tea leaves. Additionally, augmentation techniques were employed, and ablation studies were conducted to identify the optimal parameters for training the models, ultimately enhancing their performance. Table 1 describes the overview of the previous work.

3. Proposed system architecture and methodology

This section details the process of implementing the proposed deep learning model, specifically highlighting the selection of the optimal network and transfer learning model for an eight-class classification problem that includes a category for healthy leaves. The initial step involved acquiring a dataset, consisting of images for each class. To enhance the dataset's suitability for deep learning, several preprocessing techniques were applied. These included resizing images, converting them to jpg format, and employing CLAHE Histogram Equalization, Adjust Brightness, Unsharp Masking, and data augmentation. Data augmentation played a crucial role in not only expanding the training dataset but also in introducing complexity, thereby enhancing the model's robustness and its ability to generalize common patterns. Four custom-built lightweight transfer learning models, such as ResNet50V2, InceptionV3, MobileNetV2, and DenseNet169, were chosen for evaluation. In the implementation phase, the models were initially applied to the preprocessed dataset without augmentation. Subsequently, the models were re-evaluated using the augmented dataset. Performance metrics were carefully assessed to gauge the impact of data augmentation on model effectiveness. An ablation study was carried out to pinpoint the best parameters for the suggested transfer learning model and the data augmentation techniques. This involved systematically removing components of the deep learning model's architecture to analyze its behavior and performance. By scrutinizing the model's response to these alterations, optimal parameters were identified, contributing to improved overall performance. The ablation study provided valuable insights into the proposed model's architecture, facilitating the selection of the most effective parameters for constructing a robust and high-performing model. The refined model, incorporating the optimal parameters, demonstrated enhanced performance, validating the importance of the ablation study in fine-tuning the model architecture. The novelty of this study lies in the integration of fine-tuned transfer learning with Grad-CAM visualization for interpretable tea leaf disease detection. Additionally, we developed a real-time Android application to make the solution accessible to farmers in resource-constrained settings. Fig. 1 describes the flowchart of methodology.

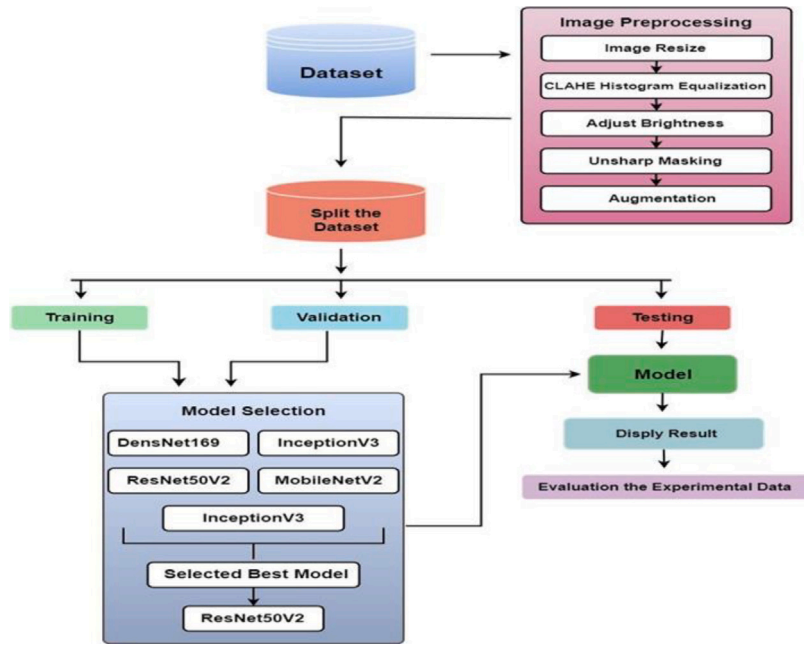


Fig. 1. Flowchart of the proposed methodology.

Table 2
Dataset description.

Disease name	Total	Training	Testing	Validation
Algal leaf	503	438	40	25
Anthracoise	145	78	37	30
Bird's eye spot	132	66	29	37
Brown blight	160	81	35	44
Gray blight	451	377	34	40
Red leaf spot	518	452	38	28
White spot	461	391	31	39
Healthy	454	398	26	30
Total	2824	2281	273	270

3.1. Dataset

The study utilized the tea dataset obtained from the open-source platform Mendeley [34]. The study's collection of images contains a depiction of the healthy leaf category, which is one of the eight categories of common diseases found on tea leaves. The seven categories are gray light (GL), white spot (WS), red leaf spot (RLS), anthracnose (ANT), brown light (BL), algal leaf spot (ALS), and bird eye spot (BES). The study involved a dataset with 2824 JPG images in total. The datasets are categorized into three parts: training, testing, and validation sections. The divided dataset is split into three sections with an 80:10:10 ratio. In the training directory there are a total of 2281 data entries, in the test directory there are 273, and in the validation directory there are 270. This dataset was categorized using pre-existing convolutional neural network structures. Fig. 2 represents the sample image of the dataset. Table 2 displays the number of images for each class.

3.2. Image preprocessing

Image pre-processing was utilized to adjust and improve the raw images prior to model training. Creating a successful model requires a careful analysis of both the network architecture and the format of the input data. Our dataset underwent pre-processing to enable the proposed model to extract pertinent features from the images. The initial phase involved normalizing and resizing the image. Also, CLAHE Histogram Equalization, brightness adjustment, and unsharp masking were applied to further preprocess the dataset. Augmentation techniques were then employed to enhance the diversity of the dataset.

3.2.1. Augmentation

Data augmentation methods were utilized to generate the final training dataset for the study. Data augmentation is the process of generating altered versions of a dataset by utilizing original data, effectively enlarging the training set. This procedure enhances

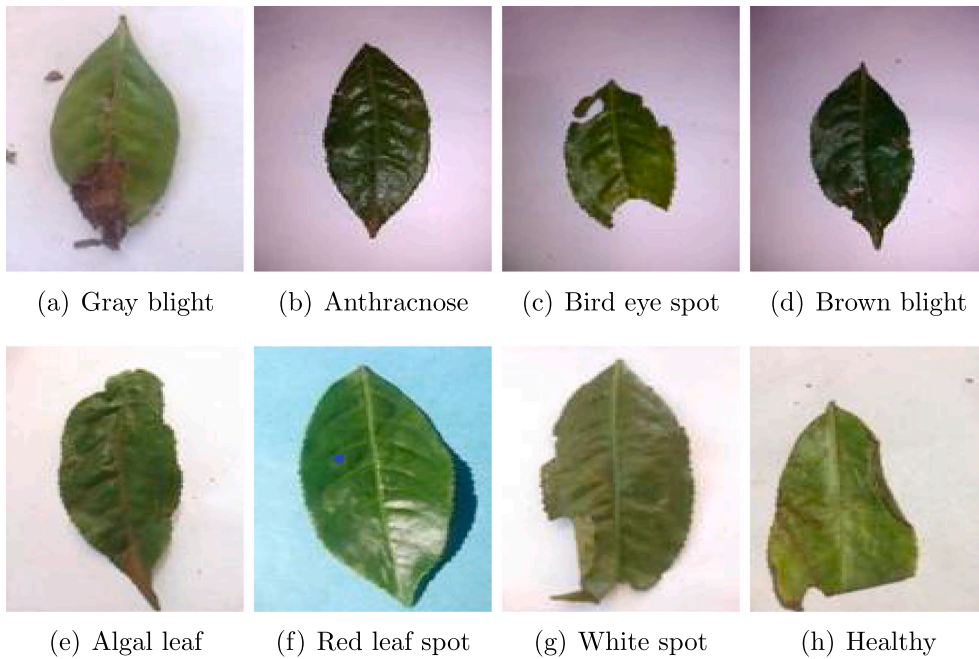


Fig. 2. Sample image of dataset.

Table 3
Image augmentation parameters for preprocessing.

Techniques	Parameter
Rescale	1.0/255
Rotation range	20
Width shift	0.2
Height shift	0.2
Shear	0.2
Zoom	0.2
Horizontal flip	True
Fill_Mode	Nearest

and widens the dataset, leading to better model performance and accuracy. Various data augmentation techniques, such as shearing, rotation, zooming, flipping, and shifting, are applied in the suggested study to boost the quantity of images in the dataset. These transformations introduced variability in orientation and perspective collectively reduced over fitting, improved generalization, and enhanced the model's resilience to real-world variability in tea leaf images. Fig. 3 represents the image of tea leaf after applying augmentation and Table 3 describes the hyper parameter for the image data augmentation.

3.2.2. Resize image

Image Resizing is the process of increasing or decreasing the size of an image. Initial preprocessing steps involved resizing images to a standardized format of 224×224 pixels. This standardization aids in creating uniformity in the dataset and ensures compatibility with deep learning models.

3.2.3. Contrast limited adaptive histogram equalization (CLAHE)

Histogram equalization is a method for shifting the intensity levels in an image to improve contrast [35]. It operates by changing the pixel intensity values in order to make the histogram of the resulting image appear nearly uniform. The original purpose of using CLAHE Histogram Equalization was to enhance the contrast of tea leaf images, which helps in better distinguishing diseased regions from healthy ones. Here we have used the clip limit value as 2 and the tile_grid_size as (14, 14).

3.2.4. Adjust brightness

Brightness adjustment was performed to normalize the lighting conditions across images, ensuring consistent input quality for the model. Changing the brightness of an image is done by adjusting the intensity values of its pixels. One easy method is to adjust each pixel by adding or subtracting a constant value [36]. To modify the brightness of an image, we alter the value of each pixel by a fixed amount. In this process the pixel value is adjusted by using a loop statement. First calculate the brightness value; if brightness

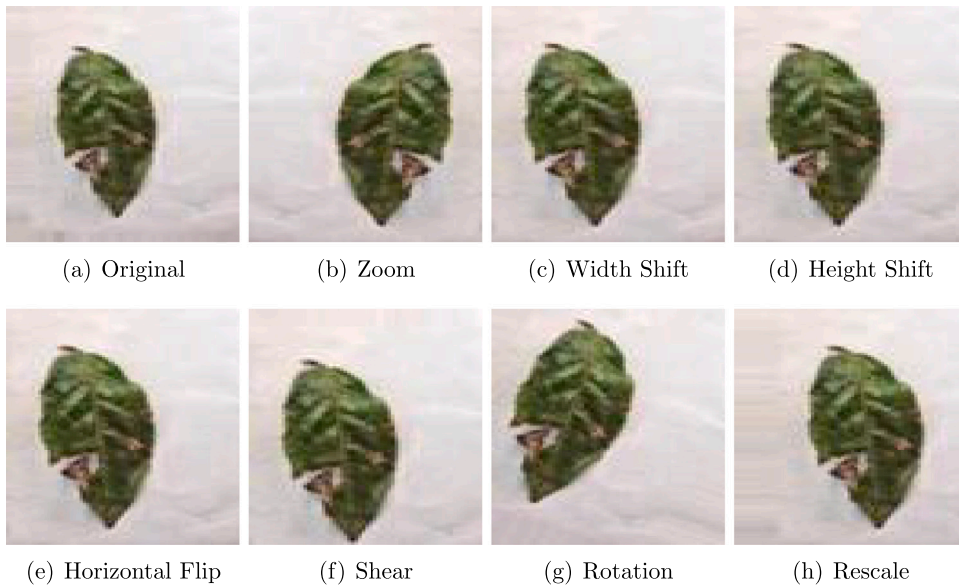


Fig. 3. Image of tea leaf after applying augmentation techniques.

Table 4

Parameters of the Image Enhancement Process.

Process name	Image enhancement value
Image resize	224 × 224 pixels
CLAHE histogram equalization	clip_limit = 2, tile_grid_size = (14, 14)
Adjust brightness	elif brightness > 120: factor = 1 elif brightness > 150: factor = 2
Unsharp making	if is_blurred: sigma = 0.3 amount = 0.3 kernel_size = (7, 7) else: sigma = 0.1 amount = 0.1 kernel_size = (5, 5)

is more than 100 then factor value is 0.5. If the value is more than 120 then the factor is 1. And finally, if the value is more than or greater than 150 then the factor value is 1.5.

3.2.5. Unsharp masking

The Unsharp Mask is designed to enhance the details in an image. It enhances images' perceived sharpness and clarity [37]. Unsharp masking is primarily used to sharpen the edges of the leaves, making it easier for the model to detect subtle disease patterns. To implement unsharp masking, the first step involves assessing whether the image is blurred or not. This information is achieved by calculating the Laplacian variance. If the Laplacian variance is less than 110, it indicates that the image is blurred. In such cases, a larger kernel size is recommended for the unsharp masking filter. Then kernel size of (7, 7) is used. On the other hand, if the image is not blurred (Laplacian variance is greater than or equal to 110), a smaller kernel size of (5, 5) is used. Fig. 4 represent the output of image preprocessing steps and Table 4 describes the parameters of the image enhancement process.

3.3. Image quality measures

Image quality measures are methods or algorithms employed to objectively evaluate the quality of an image. These metrics analyze different aspects of an image, including sharpness, contrast, noise, and overall visual perception. Different methods like MSE, RMSE, SSIM, PSNR, etc. are used to assess if an image's quality has diminished. These metrics offer information on the faithfulness and visual faithfulness of the image post-processing or compression.

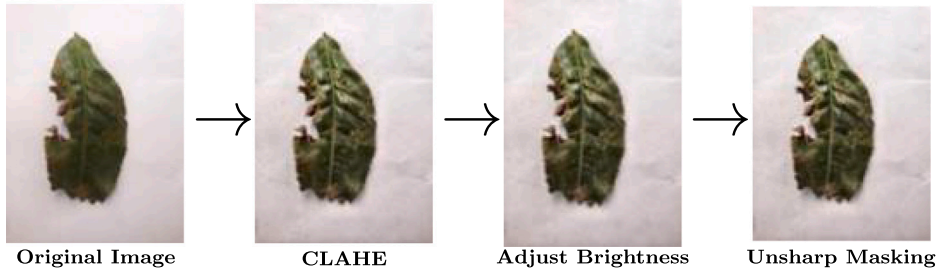


Fig. 4. Visualizing the output of image preprocessing.

3.3.1. Mean squared error (MSE)

The Mean Squared Error (MSE) is employed to assess image quality by directly measuring the average squared discrepancies among pixel values in two images [38]. This calculation aims to quantify the overall magnitude of errors between an original and a processed (or compressed) image. A lower MSE signifies enhanced similarity and higher image quality. Achieving an MSE of 0 signifies a perfect match between the images, indicating no errors. When the Mean Squared Error value increases, it indicates a growing difference between the initial and edited pictures. Therefore, a lower MSE is preferred, with 0 as the best value, representing minimal difference. The formula for it is

$$MSE = \frac{1}{N} \sum_{i=1}^N (l_i - k_i)^2 \quad (1)$$

In this case, N represents the overall number of pixels where l_i is the original image pixel values and k_i is the preprocessed image pixel values.

3.3.2. Root mean squared error (RMSE)

Root Mean Squared Error (RMSE) is commonly utilized in different fields such as image processing and signal analysis processing [39]. It is calculated from the Mean Squared Error (MSE) and provides a measure of the average magnitude of the errors between predicted and actual values.

$$RMSE = \sqrt{MSE} \quad (2)$$

3.3.3. Peak signal to noise ratio (PSNR)

PSNR is frequently utilized to measure the proportion between an image's highest potential power and the power of the signal that causes noise. It provides a straightforward and well-liked method for assessing image quality and is expressed in decibels (dB) [40]. The calculation involves computing the Mean Squared Error (MSE) between the unaltered and modified signals. SNR is a frequently employed measurement that compares the power of an image's highest potential with the power of the signal causing noise. It offers a simple and commonly used way to evaluate image quality and is measured in decibels (dB). The calculation is based on the Mean Squared Error (MSE) of the original and processed signals. The formula for PSNR is as follows:

$$PSNR = 10 \cdot \log_{10} \left(\frac{(MAX)^2}{MSE} \right) \quad (3)$$

Here, MAX refers to the maximum possible pixel value.

3.3.4. Image quality measures

The SSIM metric is used to compare two images based on their structural similarities. In addition to variations in pixel level, it also takes into account brightness and distinct characteristics [41]. SSIM is especially beneficial when alternative metrics like Mean Squared Error (MSE) or Peak Signal-to-Noise Ratio (PSNR) may not provide an accurate representation of the perceived quality. Three elements make up the SSIM formula: structure, luminance and contrast. Table 5 represents the value of image quality after preprocessing.

$$SSIM = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (4)$$

The results reveal that from Table 5, none of the images' quality has worsened based on the MSE, RMSE, PSNR, or SSIM thresholds. All MSE values are less than 100, RMSE is less than 10, PSNR is larger than 30, and SSIM is greater than 0.8, indicating that the images are of acceptable quality, with minimal error and a high structural resemblance to the reference. As a result, there is no major decrease in image quality.

Table 5
Quality check of preprocessed image.

SL NO	MSE (>100)	RMSE (>10)	PSNR (<30)	SSIM (<0.8)
01	62.52	7.90	30.17	0.88
02	43.25	6.57	31.77	0.94
03	61.33	7.83	30.25	0.95
04	48.46	6.96	31.27	0.89
05	62.17	7.88	30.19	0.92
06	62.36	7.89	30.18	0.86
07	52.22	7.22	30.95	0.87
08	59.47	7.71	30.36	0.86
09	40.96	6.40	32.00	0.87
10	58.56	7.65	30.45	0.87

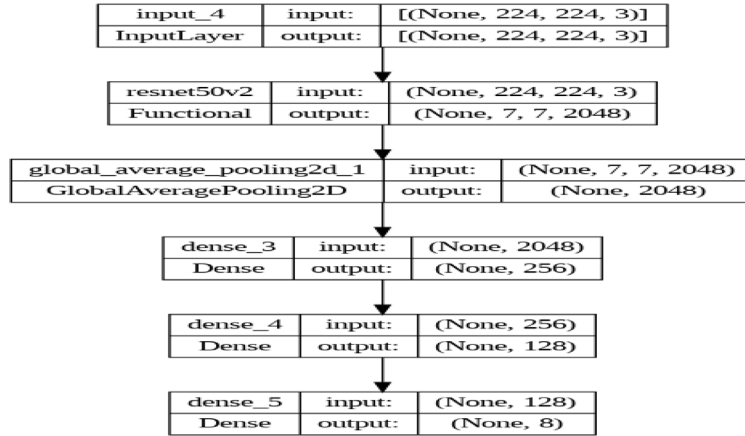


Fig. 5. Proposed model architecture.

3.4. Proposed model

The proposed model is a potent architecture for image classification tasks, leveraging the ResNet50V2 framework. ResNet50V2 was selected due to its superior performance in ImageNet classification tasks compared to ResNet50 and ResNet101 [42]. Its residual connections and optimized block formulations enable efficient training of deep networks, mitigating issues like vanishing gradients. Additionally, it also strikes a balance between performance and computational cost, making it suitable for real-time applications on resource-constrained devices. To fine-tune the model, we froze the initial layers to preserve the pre-trained weights while avoiding upsetting the learned features. This method allowed us to focus the training process on the higher layers, which were modified to reflect the distinctive features of our dataset. This method maximizes the model's performance while minimizing overfitting. The key modification in ResNet50V2 lies in the formulation of connections between blocks, resulting in improved training efficacy. The model begins with an input layer created to support images with 224×224 pixel dimensions and three RGB color channels. Afterwards, a pre-existing ResNet50V2 model functions as a strong feature extractor, employing convolutional blocks to create layered representations and generating feature maps with dimensions (None, 7, 7, 2048). To facilitate spatial dimension reduction, a Global Average Pooling 2D layer is applied after the ResNet50V2. This layer computes the average value of each feature map, resulting in a condensed tensor with dimensions (None, 2048). The following architecture has three Dense layers, each with ReLU activation functions to help in additional feature extraction and dimensionality reduction. The first Dense layer, comprising 256 neurons, enables the model to capture intricate patterns and relationships extracted from the global average-pooled features. Following this, a second Dense layer with 128 neurons continues to compress the feature representation, enhancing computational efficiency. The last Dense layer of the model contains 8 neurons and enables classification into 8 classes using SoftMax activation. The non-trainable parameters are associated with the pre-trained ResNet50V2 model. In contrast, the trainable parameters arise from the additional Dense layers, allowing the model to adapt to the specific nuances of the classification task. This integration of pre-trained and trainable components establishes a comprehensive architecture, empowering the model for robust and effective image classification. The architecture of the proposed model shows in Fig. 5.

3.5. MobileNetV2

MobileNetV2 utilizes standard residual models, implementing an inverted residual design featuring narrow bottleneck layers on both ends. Instead of enlarging the representations, it uses slim depth-wise convolutions in the middle expansion layer [43].

Freezing layers retains certain program parts unchanged during training, providing stability. The GlobalAveragePooling2D operation condenses information, capturing image highlights. Introducing fully connected dense layers enhances program understanding of complexities, with each layer acting as additional cognitive units. The last layer, utilizing a SoftMax function, finalizes decision-making. This comprehensive architecture optimizes feature extraction and decision processes for improved program performance.

3.6. Inception V3

Inception v3 is a special type of computer program for understanding pictures and finding objects in them [44]. The addition of GlobalAveragePooling2D reduces spatial dimensions, offering a global summary of features. Subsequently, three dense layers along with ReLU activation functions, facilitate feature extraction and dimensionality reduction. The final dense layer, which has been tailored with SoftMax activation, is intended for categorizing into 8 different classes. This architecture optimizes both spatial summarization and hierarchical feature learning, culminating in effective multi-class classification.

3.7. DensNet169

The DensNet-169 model belongs to the DenseNet series of models created for the purpose of image classification [45]. A fresh Sequential model is established, incorporating the pre-trained DenseNet169 as its foundation to capitalize on the acquired features. To condense spatial dimensions and derive a comprehensive feature summary, GlobalAveragePooling2D is introduced. Feature extraction and dimensionality reduction are further enhanced by the addition of two dense layers, each employing ReLU activation functions. This streamlined architecture is designed to leverage the power of transfer learning while tailoring the model for specific tasks through the custom dense layers.

3.8. Training approach

Training is a systematic process designed to guide learning toward specific outcomes, ultimately achieving performance objectives. In this model training scenario, a batch size of 64 and a maximum of 80 epochs are specified. To automate and control tasks during training epochs, callbacks are defined, with the ModelCheckpoint callback utilized for saving the model or weights. This facilitates tracking the best-performing model based on a specified metric, with a focus on monitoring validation accuracy for model evaluation. The loss function is integral, measuring the disparity between predicted and actual target values during training. Specifically, 'categorical_crossentropy' is chosen, tailored for multiclass classification with 8 output classes. The Adam optimizer is employed to update model weights based on computed gradients, adapting learning rates individually for efficiency. The optimizer is configured with an initial learning rate of 0.001, which impacts the optimization step size. The SoftMax activation function assumes a vital role, particularly for multiclass classification tasks. It transforms raw model output into probability distributions, aiding in probabilistic interpretation and stable training dynamics. This comprehensive setup orchestrates an effective training regimen for the neural network.

4. Experimental evaluation findings and analysis

4.1. Environment specification

Developing a robust experimental setting for running a research model entails several critical aspects to assure the results' reliability and validity. Here Table 6 presented the processes of experimental setup.

4.2. Evaluation metrics

A confusion matrix is a way to evaluate how accurate a classification model is in terms of its performance [46]. The display includes the number of false positives, false negatives, true positives, and true negatives.

Accuracy: One of the easiest measures to utilize is accuracy, which is frequently applied to the classification issues [47]. It shows the proportion of accurately anticipated cases to all instances. Accuracy is simple to grasp, however it might not be appropriate for datasets that are imbalanced, meaning that one class is much more than the others. The formula for accuracy is

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

The accuracy measure is calculated by dividing the number of true positive outcomes (TP) by the sum of true positive and false positive results (TP + FP). TP represents accurately predicted favorable results. The abbreviations TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively.

Precision: Precision is a measure that evaluates the accuracy of a model's predictions of positive outcomes [48]. Calculated by dividing true positives by the total of false positives and true positives. Precision is extremely important when carrying out tasks in which incorrect results could lead to significant consequences. The formula of precision is

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

Table 6
Experimental setup.

SL NO	Process name	Action
01	Input	Initiate the process by collecting tea leaf images of eight distinct classes from the open-source platform Mendeley.
02	Environment	Utilize the robust computing environment provided by Google Colab.
03	Configuration	Set up the necessary groundwork by importing all essential libraries and packages.
04	Directories configuration	Organize the workflow by loading directories dedicated to training, testing, and validation datasets.
05	Training and testing	Develop custom CNN models for transfer learning. Enhance these models by adding a fully connected layer and implementing the SoftMax activation function.
06	Model compilation	Compile the models using an Adam optimizer and a learning rate of 0.001. Implement a training regimen spanning 80 epochs, with model checkpoints using validation accuracy as a monitoring criterion.
07	Performance evaluation	Assess the models' performance comprehensively. Generate a detailed classification report, along with reports on accuracy and loss. Additionally, create a Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) graph to visualize the models' discriminative capabilities.

Table 7
The experimental result of tea leaf dataset of several models.

Model name	Accuracy	Precision	Recall	F1-Score	Training time
MobileNetV2	0.93%	0.93%	0.93%	0.93%	4500 s
DenseNet169	0.95%	0.95%	0.95%	0.95%	8260.89 s
InceptionV3	0.88%	0.89%	0.88%	0.88%	7500 s
Proposed model	0.97%	0.98%	0.98%	0.98%	6884.079 s

Recall: The recall is described as the proportion of true positives to the sum of false negatives and true positives [49]. It evaluates the model's ability to correctly recognize all relevant events. In scenarios where the absence of true positive instances is a significant issue, recall is extremely important. The formula of recall is

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

F1 Score: The F1 score represents the balanced average of recall and precision. It is particularly beneficial in cases of uneven class distribution as it finds a middle ground between recall and precision [50]. The formula for F1-Score is

$$\text{F1-Score} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (8)$$

4.3. Result analysis and deployment

4.3.1. Result analysis

This section concentrates on evaluating the effectiveness of various models employed in the study. To fully comprehend how each model behaves, accuracy and loss graphs are used to offer valuable insights into the performance trends of the models. Additionally, the ROC-AUC curve is presented for each applied model, offering a visual representation of their discriminative ability. Among the four transfer learning models — DenseNet169, ResNet50V2, MobileNetV2, and InceptionV3 — ResNet50V2 outperformed the others, achieving the highest accuracy. Specifically, MobileNetV2 attained 95% training accuracy, 90% validation accuracy, and 93% test accuracy. On the other hand, InceptionV3 showed 91% training accuracy, 86% validation accuracy, and 88% test accuracy. DenseNet169 exhibited even better results with 97% training accuracy, 94% validation accuracy, and 95% test accuracy. Optimizing the models further, a proposed model based on pre-trained fine tuning ResNet50V2 achieved remarkable results with 97% training accuracy, 94% validation accuracy, and 97% test accuracy. Overall, ResNet50V2 emerged as the most effective model, surpassing the performance of MobileNetV2, InceptionV3, and DenseNet169 across various metrics. On the other hand, the proposed model strikes a balance between accuracy and training time. It outperforms all other models in terms of accuracy and F1-score, despite taking longer to train than MobileNetV2 (4500.00 s). ResNet50V2 trains in 6884.08 s, which is faster than DenseNet169 (8260.89 s), showcasing its computational efficiency. InceptionV3, with its complex architecture, takes 7500.00 s to train but does not match the performance of the proposed model. MobileNetV2, while the fastest to train, sacrifices some accuracy. Table 7 describes the experimental result of the tea leaf dataset of several models.

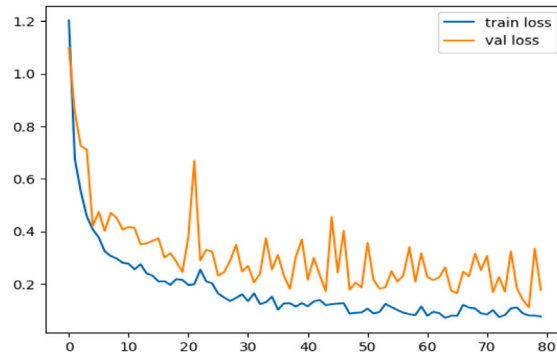


Fig. 6. Train and validation loss of proposed model.

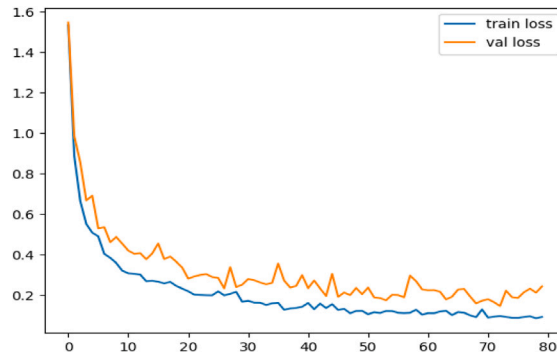


Fig. 7. Train and validation loss of DenseNet169.

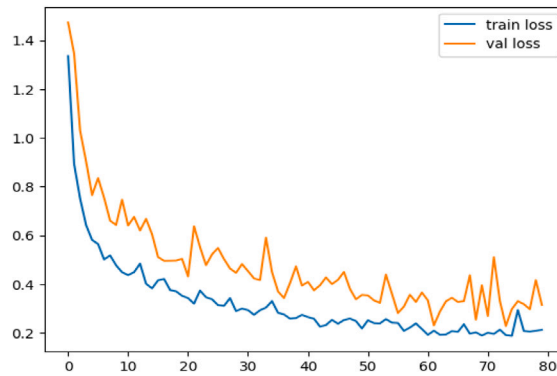


Fig. 8. Train and validation loss of MobileNetV2.

4.3.2. Train and validation loss analysis

Among the four models evaluated in this study, the proposed model demonstrated best performance by achieving the lowest train and validation losses. Figs. 6, 7, 8, and 9 depict the train and validation loss for the four distinct models

The proposed model, utilizing ResNet50V2, achieved the most favorable loss among four models. Specifically, it demonstrated a train loss of 0.0799 and a validation loss of 0.1105. In comparison, MobileNetV2 exhibited a train loss of 0.907 and a validation loss of 0.1591. DenseNet169 resulted in a train loss of 0.881 and a validation loss of 0.1594, while InceptionV3 showed a loss of 0.1899 and a validation loss of 0.2275. The proposed model displayed the superior loss metrics.

4.3.3. Train and validation accuracy analysis

The proposed model performed the best in terms of accuracy compared to the other models evaluated. Figs. 10, 11, 12, and 13 depict the train and validation accuracy for four distinct models.

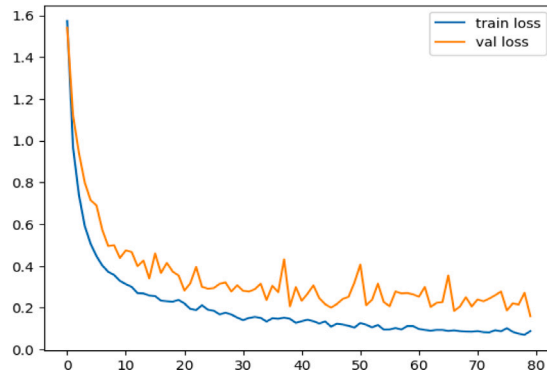


Fig. 9. Train and validation loss of InceptionV3.

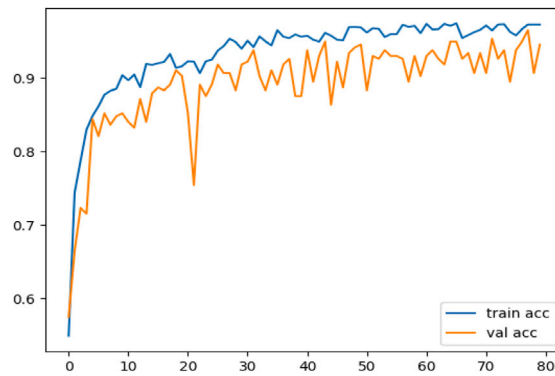


Fig. 10. Train and validation Acc of proposed model.

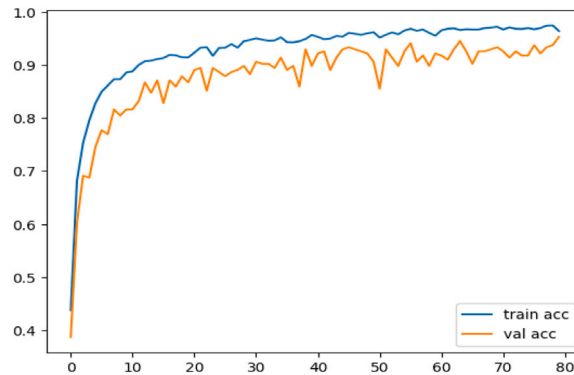


Fig. 11. Train and validation Acc of DenseNet169.

It achieved an impressive 97% accuracy in the training set and a commendable 95% accuracy in the validation set. MobileNetV2 achieved a training accuracy of 94% and a validation accuracy of 90%. The training accuracy of InceptionV3 was found to be 91%, while the validation accuracy was 86%. DenseNet169 performed better, achieving a training accuracy of 97% and a validation accuracy of 94%. The training and validation accuracy of DenseNet169 and ResNet50V2 are similar, but the proposed model outperforms them in test accuracy. In summary, the proposed model demonstrated the best training accuracy across all models evaluated.

4.4. Confusion matrix analysis

The proposed model demonstrates high accuracy across all disease types with minimal misclassifications, as shown in Fig. 15. For instance, it achieves 31 true positives for Anthracnose and 40 true positives for Brown blight, with only minor errors. Similarly,

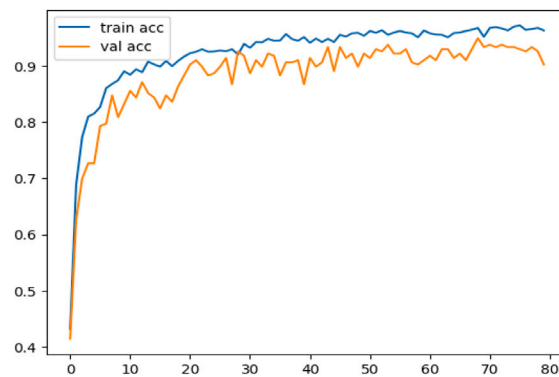


Fig. 12. Train and validation Acc of MobileNetV2.

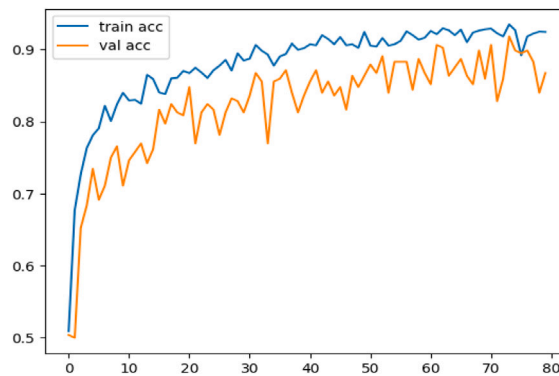


Fig. 13. Train and validation Acc of InceptionV3.

		Confusion Matrix							
True Labels	Anthracoese	31	0	0	0	0	0	0	0
	algal leaf	0	38	0	0	0	0	0	0
	bird eye spot	0	0	24	0	2	0	0	0
	brown blight	0	1	0	34	0	0	0	0
	gray_blight	0	0	0	0	40	0	0	0
	healthy	0	1	0	0	1	35	0	0
	red leaf spot	0	0	0	0	0	2	32	0
	white spot	0	1	0	0	0	0	0	28
		Anthracoese	algal leaf	bird eye spot	brown blight	gray_blight	healthy	red leaf spot	white spot
		Predicted Labels							

Fig. 14. Confusion matrix of proposed model.

DenseNet169 performs well but exhibits slightly more misclassifications compared to the proposed model, as illustrated in Fig. 16. While it attains 29 true positives for Anthracnose and 38 true positives for Gray blight, there are noticeable misclassifications, such as bird eye spot being frequently mistaken for algal leaf. Both MobileNetV2 and InceptionV3 exhibit lower accuracy and more misclassifications, particularly in distinguishing between similar disease types like algal leaf and bird eye spot. The proposed model outperformed other models due to its residual connections, which mitigate the vanishing gradient problem and enable efficient training. Its optimized block formulations enhance feature extraction and training efficacy. By freezing initial layers and modifying higher layers, the model leverages pre-trained weights while adapting to dataset nuances. In Figs. 14, 15, 16, and 17, the confusion matrices for four distinct models are shown.

It achieved an impressive 97% accuracy in the training set and a commendable 95% accuracy in the validation set. MobileNetV2 achieved a training accuracy of 94% and a validation accuracy of 90%. The training accuracy of InceptionV3 was found to be 91%, while the validation accuracy was 86%. DenseNet169 performed better, achieving a training accuracy of 97% and a validation

True Labels \ Predicted Labels	Anthracnose	algal leaf	bird eye spot	brown blight	gray_blight	healthy	red leaf spot	white spot
Anthracnose	29	0	1	0	1	0	0	0
algal leaf	0	38	0	0	0	0	0	0
bird eye spot	3	0	23	0	0	0	0	0
brown blight	0	0	0	35	0	0	0	0
gray_blight	2	0	0	0	38	0	0	0
healthy	0	0	0	1	1	35	0	0
red leaf spot	0	0	0	0	0	1	33	0
white spot	0	1	0	2	0	0	0	26

Fig. 15. Confusion matrix of DensNet169.

True Labels \ Predicted Labels	Anthracnose	algal leaf	bird eye spot	brown blight	gray_blight	healthy	red leaf spot	white spot
Anthracnose	29	0	2	0	0	0	0	0
algal leaf	0	37	0	0	0	0	0	1
bird eye spot	0	0	25	0	0	0	0	1
brown blight	0	0	0	33	2	0	0	0
gray_blight	3	0	0	0	37	0	0	0
healthy	0	1	0	0	1	33	1	1
red leaf spot	0	0	0	0	1	0	33	0
white spot	0	3	0	1	0	0	0	25

Fig. 16. Confusion matrix of MobilNetV2.

True Labels \ Predicted Labels	Anthracnose	algal leaf	bird eye spot	brown blight	gray_blight	healthy	red leaf spot	white spot
Anthracnose	30	0	0	0	1	0	0	0
algal leaf	0	37	0	1	0	0	0	0
bird eye spot	3	0	22	0	1	0	0	0
brown blight	0	0	0	33	2	0	0	0
gray_blight	4	0	0	0	35	1	0	0
healthy	0	0	0	0	1	34	2	0
red leaf spot	0	2	0	0	1	4	27	0
white spot	0	2	0	2	0	0	0	25

Fig. 17. Confusion matrix of InceptionV3.

accuracy of 94%. The training and validation accuracy of DenseNet169 and ResNet50V2 are similar, but the proposed model outperforms them in test accuracy. In summary, the proposed model demonstrated the best training accuracy across all models evaluated.

4.5. Model evaluation metrics analysis

The proposed model consistently performs well across various tea leaf disease classes, demonstrating accurate identification, particularly for Anthracnose, Brown Blight, Gray Blight, and White Spot. While other models like DenseNet169 and MobilNetV2 show competitive results in specific categories, the proposed ResNet50V2 model is a strong and reliable choice for overall disease classification. Table 8 represents the class wise accuracy, recall and precision of different models.

Table 8
Class wise Precision, Recall, and F1-Score of all models.

Model name	Class name	Precision	Recall	F1-score
Proposed model	Anthraco nose	1.00	1.00	1.00
	Algal leaf	0.93	1.00	0.96
	Bird eye spot	1.00	0.92	0.96
	Brown blight	1.00	0.97	0.99
	Gray blight	0.93	1.00	0.96
	Healthy	0.95	0.95	0.95
	Red leaf spot	1.00	0.94	0.97
	White spot	1.00	0.97	0.98
DenseNet169	Anthraco nose	0.85	0.94	0.89
	Algal leaf	0.97	1.00	0.99
	Bird eye spot	0.96	0.88	0.92
	Brown blight	0.92	1.00	0.96
	Gray blight	0.95	0.95	0.95
	Healthy	0.97	0.95	0.96
	Red leaf spot	1.00	0.97	0.99
	White spot	1.00	0.90	0.95
MobileNetV2	Anthraco nose	0.91	0.94	0.92
	Algal leaf	0.90	0.97	0.94
	Bird eye spot	0.93	0.96	0.94
	Brown blight	0.97	0.94	0.96
	Gray blight	0.90	0.93	0.91
	Healthy	1.00	0.89	0.94
	Red leaf spot	0.97	0.97	0.97
	White spot	0.89	0.86	0.88
InceptionV3	Anthraco nose	0.81	0.97	0.88
	Algal leaf	0.90	0.97	0.94
	Bird eye spot	1.00	0.85	0.92
	Brown blight	0.92	0.94	0.93
	Gray blight	0.85	0.88	0.86
	Healthy	0.87	0.92	0.89
	Red leaf spot	0.93	0.79	0.86
	White spot	1.00	0.86	0.93

4.6. Ablation study

An ablation study is a method of research that entails examining the effects of various components in a system by intentionally changing or eliminating specific parts and observing the resulting impact on overall performance. This technique is especially beneficial for assessing intricate systems like machine learning models, where it can be difficult to discern the influence of each specific element. In the context of a model like ResNet50V2, an ablation study involves examining how the removal or modification of various layers, blocks, or features influences the model's performance metrics, such as accuracy, precision, or computational efficiency. ResNet50V2 is a deep convolutional neural network known for its ability to efficiently train very deep models through residual learning. By conducting an ablation study, researchers can gain insights into how critical each layer or feature is to the model's success. To thoroughly examine the proposed model's resilience and adaptability, we conducted an ablation study, systematically adjusting key parameters to understand their impact on performance. The model achieved impressive 97% test accuracy at the beginning, using a batch size of 64, a learning rate of 0.001, and the Adam optimizer.

4.6.1. Changing the learning rate

In an experimental adjustment, the learning rate was reduced to 0.0001 while keeping the other parameters constant. This modification resulted in a test accuracy of 94%, showcasing the model's adaptability to changes in the learning rate. The [Table 9](#) shows the tried experiments with different learning rate and the best accuracy was obtained by 0.001 learning rate which is 97%.

4.6.2. Batch size modification

The batch size was adjusted from 64 to 32 while maintaining the other parameters constant. This modification led to a minor reduction in accuracy, settling at 81%. The observed stability under different batch size configurations underscores the model's resilience and its ability to maintain robust performance across varying settings. The [Table 9](#) shows the tried experiments with different batch size and the best accuracy was obtained by batch size 64 which is 97%.

4.6.3. Optimizer change

In a significant alteration, the model transitioned from the Adam optimizer to SGD. This change had a more pronounced impact on performance, resulting in an accuracy of 73%. This outcome underscores the sensitivity of the model to the optimizer choice, emphasizing the importance of optimizer selection in influencing overall accuracy. Also used Nadam optimizer but it gets 96% train

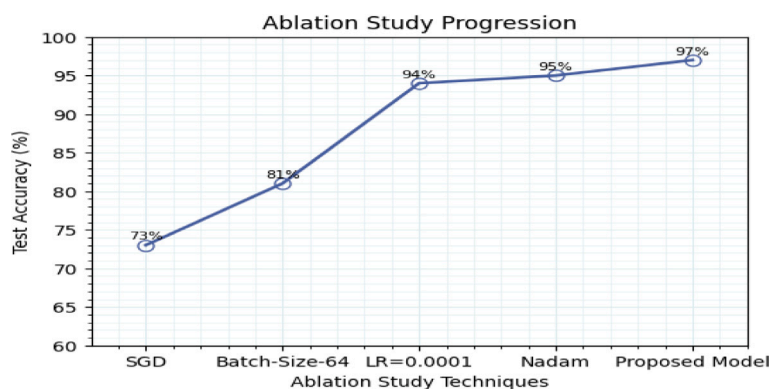


Fig. 18. Ablation study accuracy.

Table 9

Different accuracy of ablation study.

Model	Pre-processing	Batch size	Learning rate	Optimizer	Accuracy
Proposed model	With preprocessing	64	0.001	Adam	97%
	Without preprocessing	64	0.001	Adam	95%
	With preprocessing	32	0.001	Adam	96%
	With preprocessing	64	0.0001	Adam	94%
	With preprocessing	64	0.001	SGD	75%
	With preprocessing	64	0.001	Nadam	95%

Table 10

Training parameters for proposed model.

Parameter	Description
Optimization algorithm	Adam optimizer
Learning rate	0.001
Batch size	64
Number of epochs	80
Loss function	Categorical cross entropy
Activation function	ReLU
Activation function (Output layer)	SoftMax

accuracy 91% validation accuracy and 95% test accuracy. Fig. 18 describe the ablation study accuracy and table also represents the accuracy of different combination of parameter.

The original configuration of the proposed model yielded an outstanding 97% test accuracy. While minor adjustments to the learning rate and batch size resulted in slight decreases, the model consistently maintained robust performance. Notably, the shift from Adam to SGD had a more significant effect, emphasizing the model's versatility while still achieving a satisfactory overall accuracy. Despite experimenting with Nadam, it did not yield higher accuracy than the initially proposed model. Here Table 9 describes the accuracy of ablation study and Table 10 describes the best training parameter for the proposed model.

4.7. AUC and ROC curve analysis of proposed model

The model demonstrates exceptional discrimination in identifying various tea leaf diseases. Fig. 19 describes the AUC and ROC curve value of the proposed model.

It achieves flawless accuracy for Algal Leaf and Anthracnose, with very high discriminatory power in Bird Eye Spot and outstanding performance in recognizing Brown Blight and Gray Blight. The model maintains strong discrimination for Healthy, achieves high accuracy for Red Leaf Spot, and performs well for White Spot, highlighting its effectiveness in identifying positive instances and distinguishing between different tea leaf diseases.

4.8. Model confidence & class prediction analysis

The evaluation of class-wise confidence for the proposed model involves using the eight images as a representative case. The model's predictions exhibit a high level of accuracy across various plant conditions. Fig. 20 describes the confidence of the image class. In the first image, it accurately identified white spot with a confidence of 99.94%, followed by precise predictions of gray blight (99.97%) in the second image, a red leaf spot state (100%) in the third image, and gray blight (98.03%) in the fourth image.

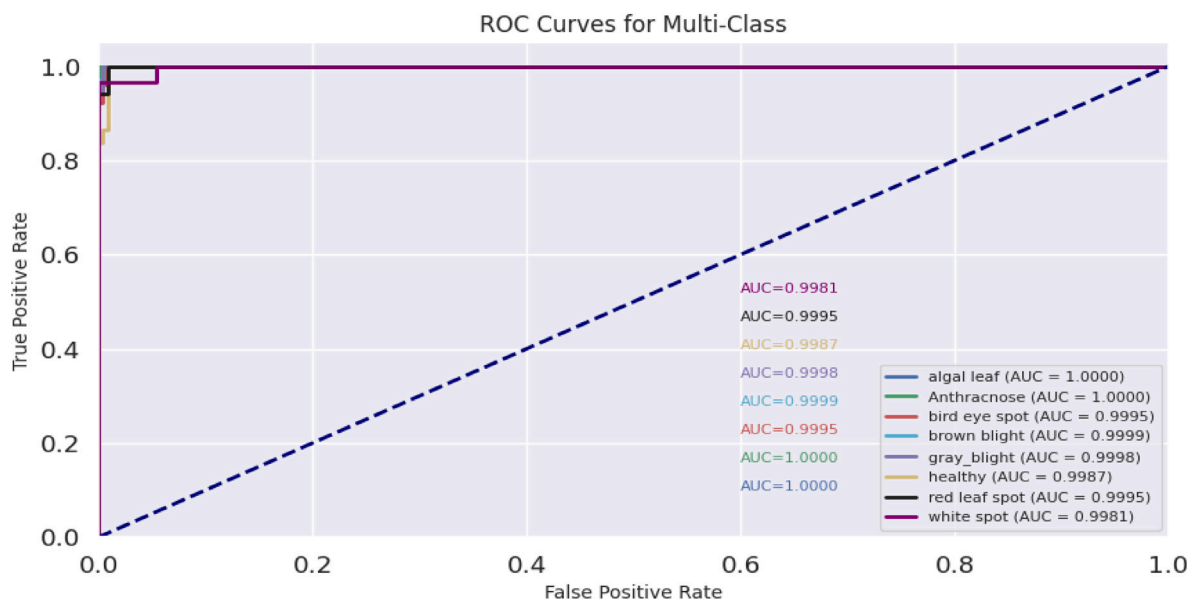


Fig. 19. AUC AND ROC curve of proposed mode.



Fig. 20. Prediction confidence of image.

The model continued its accurate streak by correctly identifying gray blight (100%) in the fifth image and bird eye spot (93.48%) in the sixth image. For the seventh and eighth images, the model correctly identified red leaf spot and white spot with confidence scores of 96.08% and 53.47%.

4.9. Gradient-weighted class activation mapping (grad-CAM) visualization

Grad-CAM, which stands for Gradient-weighted Class Activation Mapping, is a method utilized in deep learning to determine the significant areas of an image for the model's prediction. It generates a heat-map overlay on the image, highlighting regions that the model "looks at" when making its classification decision. By generating heat maps overlaying the original images, Grad-CAM highlights the regions crucial for the model's classification decisions. This enabled us to better understand how the model distinguishes between different classes of tea leaf diseases. The 'Input Image' column presents original tea leaf images classified into Algal Leaf, Anthracnose, Bird Eye Spot, and Gray Blight etc. The Grad-CAM column overlays heat maps on these images, highlighting the areas critical for disease detection as determined by the proposed model. In Table 11, Warmer colors, such as red and yellow, indicate greater significance, while cooler colors denote lesser significance. The appropriately centered heat map demonstrates that the central area of the Red Leaf Spot image has the most influence on its classification. In the case of Anthracnose, however, the right portion of the image has a higher intensity of yellow in the heat map, and the related heat map has been displayed accordingly.

Table 11
Shows the visualization of Grad-Cam image.

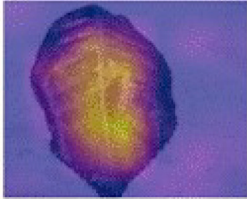
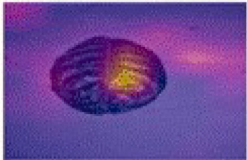
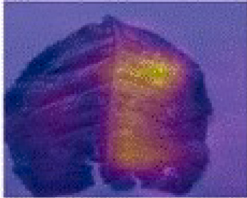
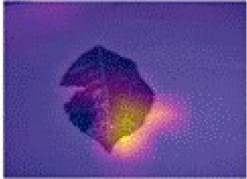
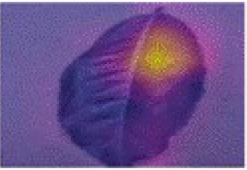
Diseased Name	Input Image	Output Image
Red Leaf Spot		
Anthracnose		
Algal Leaf		
Bird Eye Spot		
Gray Blight		
Healthy		

Table 12

Comparative analysis of the proposed model.

Author's name	Class of leaf	Algorithm	Accuracy	Limitation
Saikat Datta et al. [11]	6	CNN	96.56%	Only augmentation techniques were employed and the number of total classes is 6.
Nadide Yucel et al. [13]	8	Hybrid model	91.3%	A hybrid model was implemented which increased complexity and low accuracy.
Eliana Aida Rosyidah et al. [14]	4	CNN	95%	The achieved accuracy is comparatively low and the total number of classes is 4.
Nishmitha et al. [15]	Not Mentioned	CNN	92%	The total number of classes used remains unspecified and achieved accuracy is comparatively low.
Latha et al. [16]	8	CNN	94.45%	Only augmentation techniques were employed and accuracy is low.
Jing Chen et al. [18]	7	Leaf Net	90.23%	Accuracy is low and the number of classes is 7.
Shulin Meng et al. [19]	2	InceptionV3	95%	The number of classes used in this study is only 2 and used gray image transformation.
Jing Chen et al. [20]	7	Leaf Net	90.16%	Accuracy is low and the number of classes is 7.
Xiaoxiao et al. [21]	7	CNN	93.75%	Accuracy is low and the number of classes is 7. Also converting the colorful image into gray image transformation is applied.
Proposed model	8	ResNet50V2	97%	–

Grad-CAM visualization helped identify the regions of the tea leaf images that contributed most to the model's predictions. In the output visualization for Red Leaf Spot, the yellow regions indicate the areas that the model focused on to classify the leaf as having this disease. These highlighted areas correspond to the spots and discolorations typical of Red Leaf Spot, providing a clear indication of where the symptoms are most pronounced. Similarly, in the Anthracnose output, the yellow regions highlight specific areas of the leaf that exhibit symptoms such as dark lesions or spots, which are characteristic of this disease. For Algal Leaf, the visualization shows the model focusing on the edges and irregular patterns on the leaf, which are indicative of algal infections. Finally, in the Bird Eye Spot output, the highlighted areas point to the circular lesions that are typical of this disease, allowing for precise identification of the affected regions. These visualizations provide valuable insights into how the model interprets and classifies different leaf diseases based on their distinct symptoms. It helps in making the model's decision process more transparent. It allows researchers and practitioners to understand why the model made a particular prediction, thereby increasing trust in the model's outputs. This interpretability is crucial in applications like plant disease diagnosis, where accurate and explainable predictions are essential for effective management and treatment.

4.10. Comparative analysis

Comparative analysis involves evaluating and contrasting two or more entities to uncover similarities, differences, patterns, and trends. Table 11 describes the comparative analysis of the proposed model (see Table 12).

4.11. Deployment

Deployment is the process of deploying or implementing the model in an environment that can be accessed and used by the user for its intended purpose. The process involves selecting, training, and converting a machine learning model capable of accurately detecting tea leaf diseases into a format compatible with Android devices.

Subsequently, an Android app is developed with a user-friendly interface, allowing users to upload images of tea leaves for disease detection. The model is integrated into the app's codebase, ensuring seamless and accurate inference. An application for detecting tea leaf disease using an Android-based Smartphone has been created based on Fig. 21. In order to develop the Android app, the top-performing model was transformed into the file format "tflite" for TensorFlow lite. XML has been utilized to develop the user interface of the application. The Java programming language was used to develop both the lite TensorFlow model and the backend of the application.

5. Conclusion

Plants have a direct connection to all living organisms, particularly in terms of survival and economic concerns. Tea is considered one of the most crucial plants for individuals originating from South Asia. Tea plants suffer from various diseases like Anthracnose, Bird Eye Spot, Brown Blight, Algal Leaf, White Spot, Red Leaf Spot, and healthy. Farmers who are diagnosed early and begin treatment promptly can experience significant advantages. The goal of this research is to find solutions for each of these serious tea leaf illnesses. This research focused on the classification of diseases affecting tea leaves, utilizing a pre-trained ResNet50V2

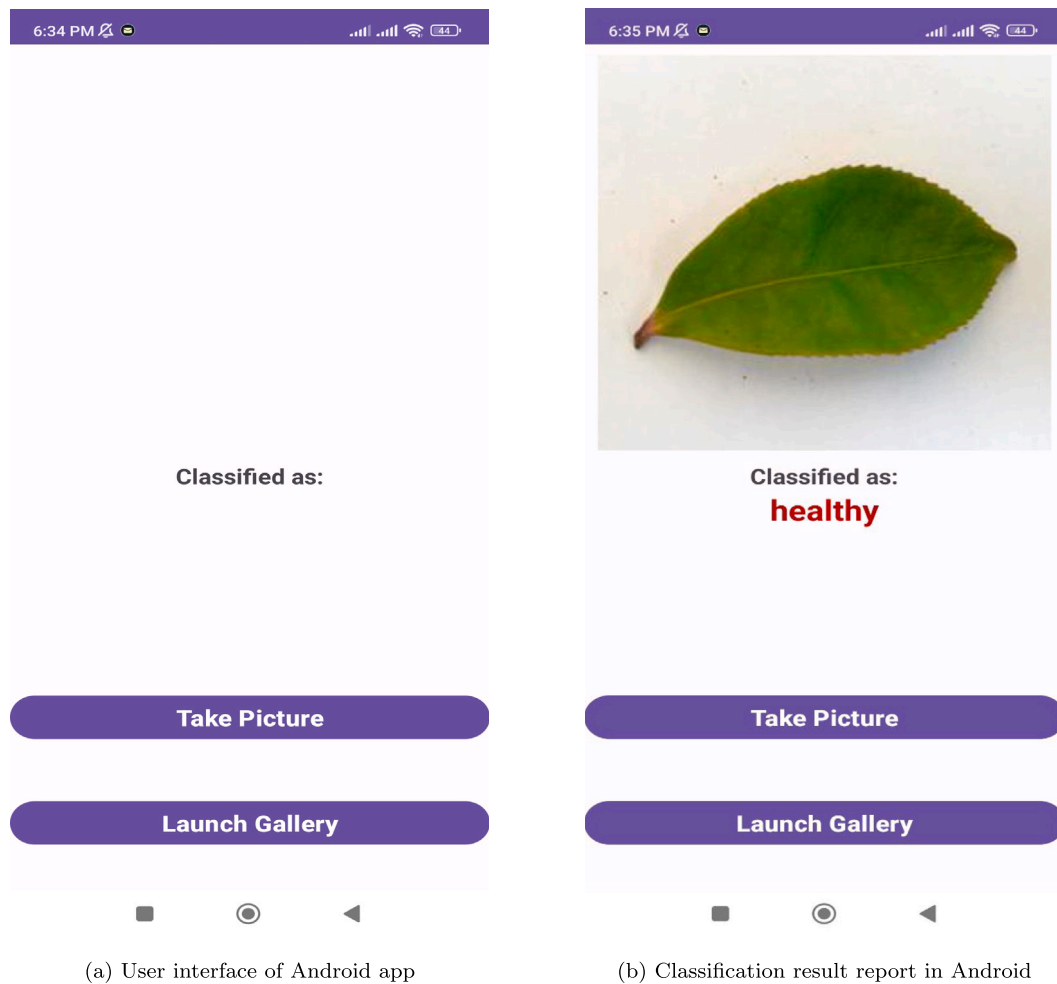


Fig. 21. Tea leaf disease classification applications.

model with fine-tuning and a specific focus on an 8-class output scenario. During the training phase, the model demonstrated promising outcomes, achieving an impressive accuracy of 97% on both the training and test sets. Different preprocessing techniques were applied to improve the model's ability to detect patterns and features in the images. These methods encompassed the use of an unsharp mask to highlight edges, adjusting contrast for improved visibility, modifying brightness for adaptability, introducing augmentation for enhanced dataset variability, and resizing for consistency in input dimensions. While the proposed model demonstrates high accuracy in classifying tea leaf diseases, certain limitations remain. The model's performance may vary when applied to other plant species or under varying image quality conditions. Future work will explore extend the model's applicability to a broader range of plant diseases.

Declaration of competing interest

The authors declare that they have no conflicts of interest to disclose. This research was conducted impartially, and there are no financial, personal, or professional relationships that could be perceived as influencing the content of this manuscript

Data availability

Data will be made available on request.

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